

Sri Lanka Institute of Advanced Technological Education
Advanced Technological Institute – Gampaha

2nd Year Individual Project

Test Cases



**Development of a Web-Based Learning Management System (LMS) For
Aura Pen Saku Education Institute**

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Module 01 – Authentication & Login Module

Test Case ID: TC_LOGIN_01

Title: Verify successful Admin login with valid credentials

Module: Authentication & Login Module

Priority: High

Preconditions: Admin user is registered in the system. Login page is accessible.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to the login page	Login page is displayed
2	Enter valid Admin username	Username is accepted
3	Enter valid Admin password	Password is accepted
4	Click on the "Login" button	User is redirected to the Admin dashboard

Test Data:

Username	Password
Admin	Admin@123

- **Expected Result:** Admin is logged in and redirected to the dashboard.
- **Actual Result:** Admin is logged in and redirected to the dashboard.
- **Status:** Pass
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_LOGIN_02

Title: Verify Teacher login functionality

Module: Authentication & Login Module

Priority: High

Preconditions: A teacher account has been assigned a username and password.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to login page	Login screen loads
2	Enter Teacher username & password	Credentials accepted
3	Click "Login"	Redirected to Teacher Dashboard

Test Data:

Username	Password
Buddhika	TC@123

- **Expected Result:** Teacher is logged in and redirected to the Teacher Dashboard.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_LOGIN_03

Title: Verify Student login functionality

Module: Authentication & Login Module

Priority: High

Preconditions: A student account has been registered in the system.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to login page	Login screen loads
2	Enter Student username & password	Credentials accepted
3	Click "Login"	Redirected to Student Dashboard

Test Data:

Username	Password
Kamal	Stu@123

- **Expected Result:** Student is logged in and redirected to the Student Dashboard.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_LOGIN_04

Title: Verify login with invalid password

Module: Authentication & Login Module

Priority: High

Preconditions: Admin user is registered in the system. Login page is accessible.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to the login page	Login page is displayed
2	Enter valid username	Username is accepted
3	Enter invalid password	Password is accepted in the field
4	Click on the "Login" button	System shows an error message

Test Data:

Username	Password
Admin	WrongPass123

- **Expected Result:** System displays an "Invalid Credentials" error message.
- **Actual Result:** System displayed the error message correctly.
- **Status:** Pass
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_LOGIN_05

Title: Verify Forgot Password functionality for Students/Teachers

Module: Authentication & Login Module

Priority: Medium

Preconditions: User (Student/Teacher) has a registered email address in the system.

Test Steps:

Step No.	Action	Expected Result
1	Click "Forgot Password" link	Redirected to password recovery page
2	Enter registered student/teacher email	Email is accepted
3	Click "Send Reset Link"	System confirms email sent

Test Data:

Email Address
student@gmail.com

- **Expected Result:** A password reset link is successfully sent to the student's/teacher's email.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_LOGOUT_01

Title: Verify successful Logout functionality

Module: Authentication & Login Module

Priority: High

Preconditions: User is currently logged into the system.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to the top navigation bar	Profile/Logout options are visible
2	Click on the "Logout" button	System processes the logout request
3	Try to access the dashboard via Back button	Access is denied

Test Data:

Action Type
Button Click

- **Expected Result:** User is logged out and safely redirected to the login interface.
- **Actual Result:** User was successfully logged out and redirected.
- **Status:** Pass
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Module 02 – Admin Dashboard Module

Test Case ID: TC_ADMIN_01

Title: Verify adding a new teacher with valid details

Module: Admin Dashboard Module

Priority: High

Preconditions: User is logged in as Admin. "Manage Teachers" page is accessible.

Test Steps:

Step No.	Action	Expected Result
1	Click on "+ Add Teacher" button	Teacher registration form opens
2	Fill in all required fields	Data is accepted
3	Click the "Save" button	Details are saved in the database

Test Data:

Name	Subject	Phone
Charaka Dikwella	Science	070 2303350

- **Expected Result:** The new teacher is saved and appears in the Manage Teachers table.
- **Actual Result:** The teacher was saved and displayed in the table correctly.
- **Status:** Pass
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_ADMIN_02

Title: Verify editing existing teacher details

Module: Admin Dashboard Module

Priority: Medium

Preconditions: User is logged in as Admin. At least one teacher exists in the system.

Test Steps:

Step No.	Action	Expected Result
1	Click "Edit" on a teacher record	Edit form opens with current details
2	Modify the phone number	New data is accepted
3	Click "Update"	Details are updated in the database

Test Data:

Target Teacher	New Phone
Charaka Dikwella	071 9998887

- **Expected Result:** Updated details are saved and successfully reflected in the table.
- **Actual Result:** Changes were saved and displayed successfully.
- **Status:** Pass
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_ADMIN_03

Title: Verify deleting a teacher from the system

Module: Admin Dashboard Module

Priority: High

Preconditions: User is logged in as Admin. Teacher record exists in the table.

Test Steps:

Step No.	Action	Expected Result
1	Click the "Delete" icon on a record	Confirmation prompt appears
2	Click "Confirm" or "Yes"	Record is deleted from the database

Test Data:

Target Record ID / Name
Charaka Dikwella

- **Expected Result:** Teacher record is permanently removed from the system.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_ADMIN_04

Title: Verify registration of a new student

Module: Admin Dashboard Module

Priority: High

Preconditions: User is logged in as Admin. "Manage Students" page is accessible.

Test Steps:

Step No.	Action	Expected Result
1	Click "+ Register Student"	Registration form appears
2	Enter Name, Grade, Phone	Fields accept valid inputs
3	Select Subjects via checkboxes	Subjects are selected
4	Click "Save" button	Student is registered

Test Data:

Name	Grade	Subjects
Kamal Perera	Grade 9	Science

- **Expected Result:** Student is saved successfully and shown in the student list.
- **Actual Result:** Student was saved and appeared in the table.
- **Status:** Pass
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_ADMIN_05

Title: Verify updating student details

Module: Admin Dashboard Module

Priority: Medium

Preconditions: User is logged in as Admin. Student record exists.

Test Steps:

Step No.	Action	Expected Result
1	Click "Edit" on Kamal Perera	Edit form opens
2	Change Grade to Grade 10	Dropdown value changes
3	Click "Update"	Data is saved

Test Data:

Target Student	New Grade
Kamal Perera	Grade 10

- **Expected Result:** Student changes are saved and displayed correctly in the table.
- **Actual Result:** Grade was updated and displayed successfully.
- **Status:** Pass
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_ADMIN_06

Title: Verify deleting a student record

Module: Admin Dashboard Module

Priority: High

Preconditions: User is logged in as Admin. Student record exists.

Test Steps:

Step No.	Action	Expected Result
1	Locate student record	Record is visible
2	Click "Delete" button	Confirmation box appears
3	Confirm deletion	Record is deleted

Test Data:

Target Record
Kamal Perera

- **Expected Result:** Student record is completely removed from the table.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_ADMIN_07

Title: Verify adding a new subject

Module: Admin Dashboard Module

Priority: Medium

Preconditions: User is logged in as Admin. "Manage Subjects" page is accessible.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to "Manage Subjects"	Subject form is visible
2	Enter Subject Code and Name	Inputs are accepted
3	Click "+ Add Subject"	Subject is added

Test Data:

Subject Code	Subject Name
001	Mathematics

- **Expected Result:** Subject is added and becomes available in system dropdowns.
- **Actual Result:** Subject was successfully saved and listed.
- **Status:** Pass
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_ADMIN_08

Title: Verify managing Admin Profile

Module: Admin Dashboard Module

Priority: Low

Preconditions: User is logged in as Admin.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to Admin Profile Settings	Profile page is displayed
2	Update admin email address	New email is accepted
3	Click "Save Changes"	System confirms update

Test Data:

Old Email	New Email
admin@aurapen.com	newadmin@aurapen.com

- **Expected Result:** Admin profile details are successfully updated.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Module 03 – Teacher Dashboard Module

Test Case ID: TC_TEACH_01

Title: Verify adding a class to the timetable

Module: Teacher Dashboard Module

Priority: High

Preconditions: Subjects and Teachers are already registered in the system.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to Manage Timetable	Timetable interface loads
2	Select Subject	Dropdown shows options
3	Enter Day and Time	Details are accepted
4	Click "Save Schedule"	Schedule is saved

Test Data:

Subject	Time
Mathematics	Monday 08:00 AM

- **Expected Result:** Class schedule is saved and visible on the timetable view.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_TEACH_02

Title: Verify uploading learning materials

Module: Teacher Dashboard Module

Priority: High

Preconditions: User is logged in as Teacher.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to "Learning Materials"	Upload interface loads
2	Select a PDF file	File is attached
3	Click "Upload"	System processes the file

Test Data:

File Name	Subject
Chapter1.pdf	Mathematics

- **Expected Result:** File is uploaded successfully and visible to assigned students.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_TEACH_03

Title: Verify viewing enrolled students for a subject

Module: Teacher Dashboard Module

Priority: Medium

Preconditions: User is logged in as Teacher.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to "My Students"	Interface loads
2	Select Subject from dropdown	Selection is accepted
3	Click "View List"	Table populates

Test Data:

Target Subject
Mathematics

- **Expected Result:** A list of all students enrolled in the selected subject is displayed.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Module 04 – Student Dashboard Module

Test Case ID: TC_STUD_01

Title: Verify URL protection against unauthorized Admin access

Module: Student Dashboard Module

Priority: High

Preconditions: User is logged in as a "Student" (Not Admin).

Test Steps:

Step No.	Action	Expected Result
1	Login using Student credentials	Student dashboard loads
2	Manually type Admin URL	URL is entered
3	Press Enter	System restricts access

Test Data:

URL to test
http://localhost:3000/manage-teachers

- **Expected Result:** Student is blocked from viewing the Admin page and shown "Access Denied".
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_STUD_02

Title: Verify viewing student timetable

Module: Student Dashboard Module

Priority: High

Preconditions: User is logged in as Student. Timetable is already created by the Teacher.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to "My Timetable"	Timetable page loads
2	Check scheduled classes	Classes match registered subjects

Test Data:

Registered Subject
Mathematics

- **Expected Result:** Student can successfully view their specific class schedule.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25

Test Case ID: TC_STUD_03

Title: Verify downloading learning materials

Module: Student Dashboard Module

Priority: High

Preconditions: User is logged in as Student. Materials have been uploaded by the Teacher.

Test Steps:

Step No.	Action	Expected Result
1	Navigate to "Learning Materials"	Materials list loads
2	Locate specific file	File is visible
3	Click "Download" icon	Browser starts downloading

Test Data:

Target File
Chapter1.pdf

- **Expected Result:** File is successfully downloaded to the student's device.
- **Actual Result:** (To be filled after test execution)
- **Status:** Not Executed
- **Tested By:** Koshila Madushani
- **Date:** 2026-06-25